

Worksheet — Linear Equations (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions in your notebook. Show every step of your working for the subjective questions; choose the single best option for MCQs. Verify each solution by substitution. Total time: 50 minutes.

Q1. The standard form of a linear equation in one variable is: [1 mark]

- (A) $ax^2 + bx + c = 0$
- (B) $ax + b = 0$ ($a \neq 0$)
- (C) $ax + by + c = 0$
- (D) $x = c$

Q2. A linear equation in two variables has exactly one solution. [1 mark]

- (A) True
- (B) False

Q3. The unique solution of $ax + b = 0$ ($a \neq 0$) is $x = \underline{\hspace{2cm}}$. [1 mark]

Q4. Solve $5x - 3 = 2x + 9$. [1 mark]

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 6$

Q5. The system $2x + 3y = 7$, $6x + 9y = 21$ has: [2 marks]

- (A) A unique solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Exactly two solutions

Q6. Solve $3(x - 2) + 4 = 2x + 1$. Show every step and verify. [3 marks]

Q7. Solve $x/2 + (x + 1)/3 = 4$. [3 marks]

Q8. The sum of three consecutive odd integers is 75. Find the integers. [3 marks]

Q9. Solve by substitution: $x + y = 7$, $x - y = 1$. [3 marks]

Q10. Five pencils and three erasers cost ■40; three pencils and two erasers cost ■25. Find the price of each. [6 marks]

Q11. A motorboat travels 24 km downstream in 3 hours and the same distance upstream in 4 hours. Find the speed of the boat in still water and the speed of the current. [6 marks]

Answer Key

Worksheet — Linear Equations (Class 10 NIOS Mathematics)

Q1. The standard form of a linear equation in one variable is:

Answer: $ax + b = 0$ ($a \neq 0$)

Linear means the variable appears only to the first power; standard form is $ax + b = 0$ with $a \neq 0$.

MCQ · EASY · 10-NIOS-MATH-LE-01

Q2. A linear equation in two variables has exactly one solution.

Answer: False

It has infinitely many solutions — every point on the line.

TRUE_FALSE · EASY · 10-NIOS-MATH-LE-04

Q3. The unique solution of $ax + b = 0$ ($a \neq 0$) is $x = \underline{\hspace{1cm}}$.

Answer: $-b/a$

Subtract b , divide by a .

FILL_BLANK · EASY · 10-NIOS-MATH-LE-01

Q4. Solve $5x - 3 = 2x + 9$.

Answer: $x = 4$

$5x - 2x = 9 + 3 \rightarrow 3x = 12 \rightarrow x = 4$.

MCQ · EASY · 10-NIOS-MATH-LE-02

Q5. The system $2x + 3y = 7$, $6x + 9y = 21$ has:

Answer: Infinitely many solutions

$2/6 = 3/9 = 7/21 = 1/3$ — same line.

MCQ · MEDIUM · 10-NIOS-MATH-LE-05

Q6. Solve $3(x - 2) + 4 = 2x + 1$. Show every step and verify.

Answer: Distribute: $3x - 6 + 4 = 2x + 1 \rightarrow 3x - 2 = 2x + 1 \rightarrow x = 3$. Verify: LHS = $3(3 - 2) + 4 = 7$; RHS = $2(3) + 1 = 7$ ✓.

Standard 5-step solving plan.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-LE-02

Q7. Solve $x/2 + (x + 1)/3 = 4$.

Answer: Multiply both sides by 6: $3x + 2(x + 1) = 24 \rightarrow 3x + 2x + 2 = 24 \rightarrow 5x = 22 \rightarrow x = 22/5$.

Clear denominators by the LCM, distribute, isolate.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-LE-02

Q8. The sum of three consecutive odd integers is 75. Find the integers.

Answer: Let smallest = x ; the three are x , $x + 2$, $x + 4$. Sum: $3x + 6 = 75 \rightarrow x = 23$. Integers: 23, 25, 27.

Translate, form ONE equation, solve, list back.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-LE-03

Q9. Solve by substitution: $x + y = 7$, $x - y = 1$.

**Answer: From eq.1: $y = 7 - x$. Substitute into eq.2: $x - (7 - x) = 1 \rightarrow 2x - 7 = 1 \rightarrow x = 4$; $y = 3$.
Solution (4, 3).**

Substitution is fastest when a coefficient is 1 or -1 .

SHORT_ANSWER · EASY · 10-NIOS-MATH-LE-06

Q10. Five pencils and three erasers cost ₹40; three pencils and two erasers cost ₹25. Find the price of each.

**Answer: Let pencil = x , eraser = y . $5x + 3y = 40$ and $3x + 2y = 25$. Multiply eq.1 by 2 and eq.2 by 3:
 $10x + 6y = 80$, $9x + 6y = 75$. Subtract: $x = 5$; substitute: $15 + 2y = 25 \rightarrow y = 5$. Pencil ₹5; eraser ₹5.**

Translate, eliminate y , solve.

LONG_ANSWER · MEDIUM · 10-NIOS-MATH-LE-08

Q11. A motorboat travels 24 km downstream in 3 hours and the same distance upstream in 4 hours. Find the speed of the boat in still water and the speed of the current.

Answer: Let still-water speed = x km/h, current = y km/h. Downstream: $x + y = 24/3 = 8$. Upstream: $x - y = 24/4 = 6$. Add: $2x = 14 \rightarrow x = 7$; subtract: $2y = 2 \rightarrow y = 1$. Boat speed 7 km/h; current 1 km/h.

Set up two linear equations from downstream / upstream times.

LONG_ANSWER · HARD · 10-NIOS-MATH-LE-08